

Turbomachinery Compressor Design

(Part of the T-AXI suite of codes)

Version 1.1

USER MANUAL

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Mark G. Turner

Department of Aerospace Engineering & Engineering Mechanics
University of Cincinnati, OH, USA
mark.turner@uc.edu

Ali Merchant

Gas Turbine Laboratory
Massachusetts Institute of Technology, MA, USA
merchant@mit.edu

Dario Bruna

Department of Fluid-Machinery, Energy Systems and Transportation
University of Genoa, Italy
dario.bruna@unige.it

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Mark G. Turner, Ali Merchant, Dario Bruna

Introduction

The T-C_DES (Turbomachinery Compressor DESign) code is a meanline axial flow compressor design tool, the first step of T-AXI, an axisymmetric method for a complete turbomachinery geometry design. This manual describes the thermodynamic theory of the code and the I/O files.

Nomenclature

a	Speed of sound	H	Total enthalpy
c	chord	M	Mach number
h	Static enthalpy or blade height	P	Pressure
$\dot{m}$	Mass flow	R_{gas}	Gas constant
q	Meridional velocity magnitude	S	Blade spacing, $S = 2\pi r / N_{blades}$
r	Radius	T	Temperature
s	Entropy	U	Rotor wheel speed
A	Area	V	Absolute velocity
C_p	Specific heat at constant pressure	W	Rotor relative velocity
DF	Diffusion factor		
DR	Diffusion ratio		

Greek

α	Absolute flow angle	θ	Tangential direction
β	Relative flow angle	ρ	Density
δ	Blockage source	σ	Solidity
δ^*	Boundary layer displacement thickness	Ω	Rotor angular velocity
γ	Ratio of specific heats	$\bar{\omega}$	Compressor loss coefficient
λ	Blockage coefficient		

Subscripts

h	Hub	S	Secondary flow
LE	Leading Edge	T	Total
m	Mean or pitch	TE	Trailing Edge
t	Tip	W	Work
x	Axial direction	θ	Tangential direction
Q	Heat Transfer		

Stage Equations – Rotor & Stator + IGV

This section of the manual deals about the theory and the equations included in the code and used for the turbomachinery design. The basic configuration for an axial-flow compressor often comprises the IGV (*inlet guide vane*) and a series of *stages*, combination of a rotating row (*rotor*) and a stationary row (*stator*). The last stage usually has a guide vane (or stator) that turns the fluid into the axial direction.

IGV	The first blade row: it develops the swirl or tangential velocity for which the first rotating row is designed. Guide vanes also serve the purpose of preventing the injection of foreign objects into the compressor and of breaking up large turbulent eddies in the atmosphere. The flow is accelerated through the IGV, resulting in a decrease in static pressure. If the first rotating row is designed for no inlet swirl, the inlet guide vane will normally be omitted. The IGV is a fixed vane and it is here treated as a generic stator row.
ROTOR	The rotor, with its rotational motion, imparts energy to the working medium: it elevates the total pressure, the temperature and the kinetic energy of the fluid converting the major part of the mechanical energy into potential energy.
STATOR	The stator row removes the swirl developed by the rotor to convert kinetic energy to static pressure (by flow diffusion) and to establish the proper swirl velocity for the flow to enter in the next rotor.

The inlet and the outlet sections of each stage are labeled in this way:

0. *Inlet IGV*
1. Inlet Rotor = *Outlet IGV*
2. Outlet Rotor = Inlet Stator
3. Outlet Stator = *Inlet Rotor for Multistage compressor*

As already mentioned, the rotor is the rotational element by which it is possible to transfer energy to the fluid. Usually its flow is viewed and analyzed in the *relative system* (also rotating frame), fixed to the rotor geometry. The *absolute system* (also stationary frame) is the reference, fixed to the ground, used to study the stator rows. The relationship between the relative velocity vector $\vec{W}$ and the absolute velocity vector $\vec{V}$ is:

$$\vec{V} = \vec{U} + \vec{W}$$

where $\vec{U}$ is the *rotor wheel speed*,

$$\vec{U} = \Omega \times r$$

Ω is the *rotor angular velocity* and r is the meanline radius in the selected section. In the T-AXI suite the rotation direction of Ω is defined positive if it is clockwise when the observer, behind the blade row, is looking forward.

The next figure (Fig.1) shows the velocity triangles for a complete axial compressor stage, including the IGV row at the inlet and the second rotor row at the outlet section.

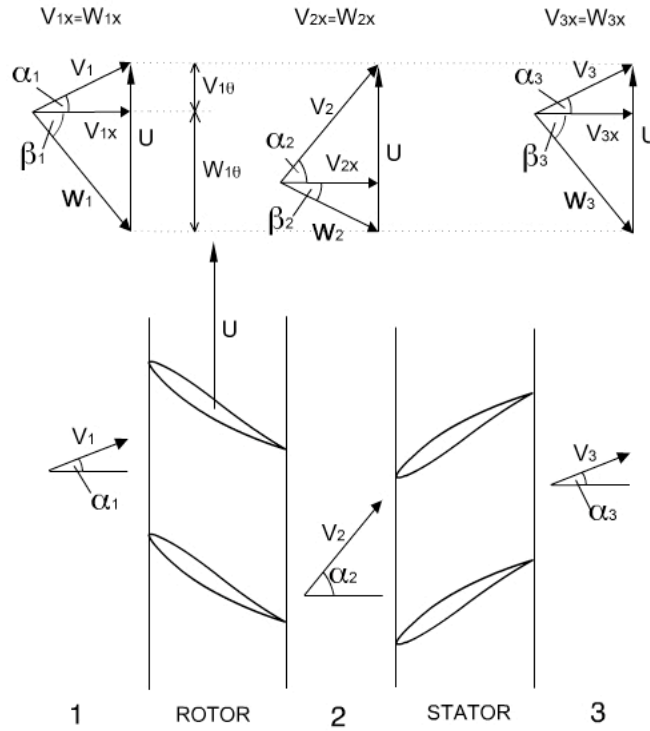


Fig. 1 – Velocity triangles diagram

These are the velocity triangle relations:

$$\alpha = \tan^{-1}\left(\frac{V_{\theta}}{V_x}\right) ; V = \frac{V_x}{\cos \alpha}$$

$$W_{\theta} = V_{\theta} - U ; W_x = V_x ; \beta = \tan^{-1}\left(\frac{W_{\theta}}{W_x}\right) W = \frac{W_x}{\cos \beta}$$

Euler's Turbomachinery equation is used to define the total enthalpy rise due to stage work (transferred by the rotor row) as:

$$\Delta H_w = \int \Omega d(rV_{\theta})$$

This equation can be applied across the rotor:

$$H_2 - H_1 = C_p(T_{T2} - T_{T1}) = \Omega(r_2 \overline{V_{2\theta}} - r_1 \overline{V_{1\theta}})$$

The free vortex assumption is applied where

$$rV_{\theta} = const$$

across the span of the blade row.

The following equations are some of all the formulations included in the code. Starting from the data included in the input files, using these or rearranging these formulations, it is possible to calculate the flow and the geometric quantities showed in the output files and needed for the design.

$$\frac{T}{T_t} = \frac{1}{1 + \left[\frac{(\gamma-1)}{2} \right] M^2} ; \quad P = \frac{P_t}{\left\{ 1 + \left[\frac{(\gamma-1)}{2} \right] M^2 \right\}^{\frac{\gamma}{\gamma-1}}}$$

$$a = \sqrt{\gamma R_{gas} T} ; \quad V = Ma$$

$$\rho = \frac{P}{R_{gas} T} ; \quad A = \frac{\dot{m}}{\rho V_x \lambda}$$

In reality the *blockage coefficient* λ is a term that represents the area reduction due to endwall boundary layers, wakes, and other circumferential non-uniformities. For the T-AXI solver with the coupled boundary layer turned on, some of this blockage affect will be calculated.

The cross sectional area is then:

$$A = \pi (r_t^2 - r_h^2)$$

A hub, tip or midspan radius is given. From this and the area, the flowpath radii can be defined. It should be realized that the design aspect of turbomachinery is to determine how the flowpath area varies. This is the real output of this design code, and is the critical quantity in an optimum compressor design.

The loss coefficient is specified as an input. The total pressures in the following equation are relative total pressure if applied for a rotor:

$$\bar{\omega} = \frac{P_{T2} - P_{T1}}{P_{T1} - P_1}$$

The average loss coefficient is also calculated in the T-AXI solver. The input loss coefficient can be updated, or viewed as a parameter that is used along with the other parameters to create the flowpath area of the compressor. If the loss is not updated, the other specified parameters such as the Mach number and angles will not be set precisely, but will vary depending on the actual loss calculated.

<i>Compressor Pressure Ratio</i>	$PR_{Comp} = \frac{P_{Tout}}{P_{Tin}}$
<i>Compressor Temperature Ratio</i>	$TR_{Comp} = \frac{T_{Tout}}{T_{Tin}}$
<i>Compressor Adiabatic Efficiency</i>	$\eta_{Comp_ad} = \frac{PR_{Comp}^{\frac{\gamma-1}{\gamma}} - 1}{TR_{Comp} - 1}$
<i>Compressor Polytropic Efficiency</i>	$\eta_{Comp_pol} = \frac{\gamma - 1}{\gamma} \frac{\ln(PR_{Comp})}{\ln(TR_{Comp})}$

Table 1 – Compressor performance parameters

In Table 1 the performance parameters for the complete axial flow compressor are listed. If the design includes the IGV two different values for the *Compressor Pressure Ratio* will be calculated:

1. (No IGV) - $PR_{Comp} = \frac{P_{T_{out}}}{P_{T_{in}}} \quad P_{T_{in}} = P_T @ \text{Section 1, 1st Stage}$
2. (With IGV) - $PR_{Comp} = \frac{P_{T_{out}}}{P_{T_{in}}} \quad P_{T_{in}} = P_T @ \text{Section 0, 1st Stage}$

The definitions of the stage properties (included in the *Tcdes-results* file) are listed in Table 2.

<i>Stage Pressure Ratio</i>	$PR = \frac{P_{T3}}{P_{T1}}$
<i>Stage Adiabatic Efficiency</i>	$\eta_{ad} = \frac{PR^{\frac{\gamma-1}{\gamma}} - 1}{\frac{T_{T3}}{T_{T1}} - 1}$
<i>Stage Polytropic Efficiency</i>	$\eta_{pol} = \frac{\gamma - 1}{\gamma} \frac{\ln(PR)}{\ln\left(\frac{T_{T3}}{T_{T1}}\right)}$
<i>Degree of Reaction</i>	$R = \frac{T_2 - T_1}{T_3 - T_1}$
<i>Rotor Diffusion Factor</i>	$DF_R = 1 - \frac{W_2}{W_1} \frac{ W_{1\theta} - W_{2\theta} }{2\sigma W_1}$
<i>Stator Diffusion Factor</i>	$DF_S = 1 - \frac{V_3}{V_2} \frac{ V_{2\theta} - V_{3\theta} }{2\sigma V_2}$
<i>Stage Load Coefficient</i>	$\psi = \frac{C_p(T_{T2} - T_{T1})}{U_2^2}$
<i>Stage Flow Coefficient</i>	$\phi = \frac{V_{1x}}{U_2}$

Table 2 – Stage parameters

Block Diagram – I/O

The input to T-C_DES is an “init” file, a “stage” file, and an optional “igv” file. If the “igv” file does not exist, it is assumed that there is no igv in the design. These files contain all the information needed for the design: geometrical, fluid and turbomachinery working data. Moreover in the “init” file it is also included the clearance ratio (the clearance divided by radius), an input quantity for the next axisymmetric code of T-AXI. The radius is the tip radius for a rotor and hub radius for a stator. For a preliminary compressor layout, the one clearance to radius number was felt to be more “constant” than a clearance to height or clearance to chord ratio.

The output of T-C_DES is a “tcdes-results” file, a “cgeo” file, a “crvth” file, a “stack” file and a “walls” file. Some of these files are the input for the T-AXI code and they are detailed described in this section.

Figure 2 shows the T-C_DES input/output block diagram. The file extension, here “xxx”, has to be the same for all the input files in order to run the code. Then it will return the output files named with the same extension.

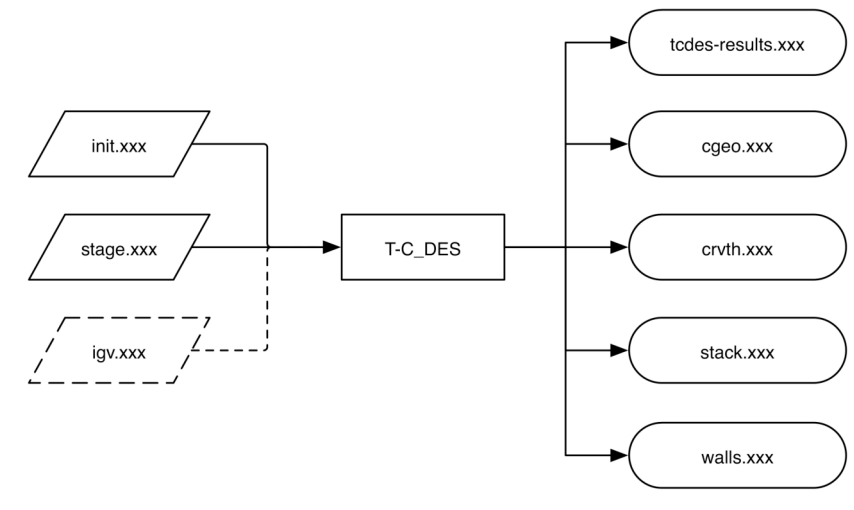


Fig. 2 – T-C_DES input/output block diagram

- **Init.xxx** – This file contains the general informations for the single or multistage compressor design. The next table shows all the options and the data included.

Units	1- SI: International System 2- EN: English System
Radius Option	1- Constant HUB 2- Constant PITCH 3- Constant TIP 4- Arbitrary
Design Option	1- Free Vortex 2- Forced Vortex
inFLAG4	
inFLAG5	
inFLAG6	
N° Stages	Maximum stage number: 20
Mass Flow Rate	SI: [kg/s] – EN: [lbm/s]
Inlet Total Pressure	SI: [Pa] – EN: [psi]
Inlet Total Temperature	SI: [K] – EN: [R]

Alpha 3 (Absolute angle) – Last Stage	[deg]
Mach 3 – Last Stage	[-]
Ratio of Specific Heats	[-]
Gas Constant	SI: [kJ/(kg K)]- EN: [ft ² /(s ² R)]
Ratio of Clearance	[-]
Radius if constant hub, mean or tip radii design option used	SI: [m] – EN: [in]

Table 3 – Init.xxx file data

- **Stage.xxx** – In this file the geometrical information and thermo-fluid dynamic data for each stage are included. The data are organized in columns whose number corresponds to the stage number of the multistage compressor. The Table 4 is an example of all the information included for each stage.

Alpha 1 (Absolute angle)	[deg]
Mach 1	[-]
Total Temperature Rise	SI: [K] – EN:
Rotor Loss Coefficient	[-]
Stator Loss Coefficient	[-]
Rotor Solidity	[-]
Stator Solidity	[-]
Stage Lambda	[-]
Stage Bleed	[-]
Rotor Aspect Ratio	[-]
Stator Aspect Ratio	[-]
Rotor Axial Velocity Ratio	[-]
Rotor Row Space Coefficient	[-]
Stator Row Space Coefficient	[-]
Stage Tip Radius	SI: [m] – EN: [in] This data has to be specified if the 4 th radius option in the “init” file is used

Table 4 – Stage.xxx file data

- **Igv.xxx** – If this file is included in the same directory with the “init” and “stage” files the design starts automatically with IGV as first blade row (Table 5).

IGV Solidity	[-]
IGV Aspect Ratio	[-]
IGV Loss Coefficient	[-]
Alpha 0 (Absolute angle)	[deg]
Mach 0	[-]
IGV Lambda	[-]
IGV Row Space Coefficient	[-]
IGV Tip Radius	SI: [m] – EN: [in]

Table 5 – Igv.xxx file data

- **Tcdes-results.xxx** – This file is a result summary for all the compressor. Many useful informations about the fluid, row and stage properties are included and summarized in the present way. These quantities are analyzed for every stage of the turbomachine in the sections:

0. *Inlet IGV*
1. Inlet Rotor = *Outlet IGV*
2. Outlet Rotor = Inlet Stator
3. Outlet Stator = *Inlet Rotor for Multistage compressor*

The thermo-fluid dynamic properties summarized in Table 6 are calculated at the different span-wise sections (hub, mean or tip).

TT	Total Temperature	SI: [K] – EN: [R]
Ts	Static Temperature	SI: [K] – EN: [R]
PT	Total Pressure	SI: [kPa] – EN: [psi]
Ps	Static Pressure	SI: [kPa] – EN: [psi]
Mach	Mach Number	[-]
Vel	Velocity	SI: [m/s] – EN: [ft/s]
Ax V	Axial Velocity	SI: [m/s] – EN: [ft/s]
TangV	Tangential Velocity	SI: [m/s] – EN: [ft/s]
α / β	Fluid Angle	SI: [m/s] – EN: [ft/s]
Radius	Streamline Radius	SI: [m] – EN: [in]

Table 6 – Thermo-fluid dynamic properties output in the “tcdes-results” file

The other quantities included in the files for each stage are:

- Section Area & Height
- Rotor Tip Relative Mach Number
- Rotor – Stator Chord & Blades Number
- Stage Pressure Ratio
- Stage Adiabatic & Polytropic Efficiency
- Degree of Reaction
- Rotor – Stator Diffusion Factor
- Stage Load & Flow Coefficient

For the complete machine:

- Compressor Pressure Ratio (With & Without IGV)
- Compressor Temperature Ratio
- Compressor Adiabatic & Polytropic Efficiency

- **cgeo.xxx** – This file presents the annulus channel geometry of the compressor: the hub and tip coordinates of the leading and trailing edge of each row are included. With this file it is possible to check the meridional view of the machine. For each stage there are four lines: two are the leading and trailing edge of the rotor, the other two the leading and trailing edge of the stator. The first and the last lines of the file are referred to the inlet and outlet sections. The first parameter is the stage index. For a design with the IGV row (two lines in the file) the IGV index is 0.

Stage Index	Hub coordinate Axis direction <i>SI: [m] – EN: [in]</i>	Hub coordinate Radius direction <i>SI: [m] – EN: [in]</i>	Tip coordinate Axis direction <i>SI: [m] – EN: [in]</i>	Tip coordinate Radius direction <i>SI: [m] – EN: [in]</i>
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Table 7 – *Cgeo.xxx* file data

- ***cvrth.xxx*** – The number of blades and the $R*V_\theta$ product are the quantities included in this file. The first parameter is the blade row index; if the IGV is present in the design (first line of the file) the first rotor data will be included in the second line.

Blade Row Index	Number of Blades	$(R_{HUB}*V_\theta)_{ROW\ INLET}$
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Table 8 – *Cvrth.xxx* file data

These last two files are the input needed to run the T-AXI solver.

- ***stack.xxx*** – This file includes the overall information for each blade row and for the full compressor axis-symmetric calculation. These are the listed data (Tab.9) for each blade row (geometry, loss, bleed and angular momentum); the geometric coordinates are non-dimensionalized by the Rotor 1 leading edge radius at the tip.

Number of Blades	Rotational Speed	Hub Clearance	Tip Clearance			
Radial_LE @Hub	Radial_TE @Hub	Axial_LE @Hub	Axial_TE @Hub	Loss	Bleed	$R_{HUB}*V_\theta$
Radial_LE @Tip	Radial_TE @Tip	Axial_LE @Tip	Axial_TE @Tip	Loss	Bleed	$R_{HUB}*V_\theta$

Table 9 – *Stack.xxx* file data

Table 10 shows the data needed for the full compressor calculation.

γ	TT_inlet [deg R]	Type Flag 0 - Comp/ 1 - Turb	Normalized Mass Flow	Re	Boundary Layer Switch	Viscous Flag
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Table 10 – *Stack.xxx* file data

- ***walls.xxx*** – The dimensionless coordinates of the meridional section of the compressor are here listed.

References

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